Functions of WBCs

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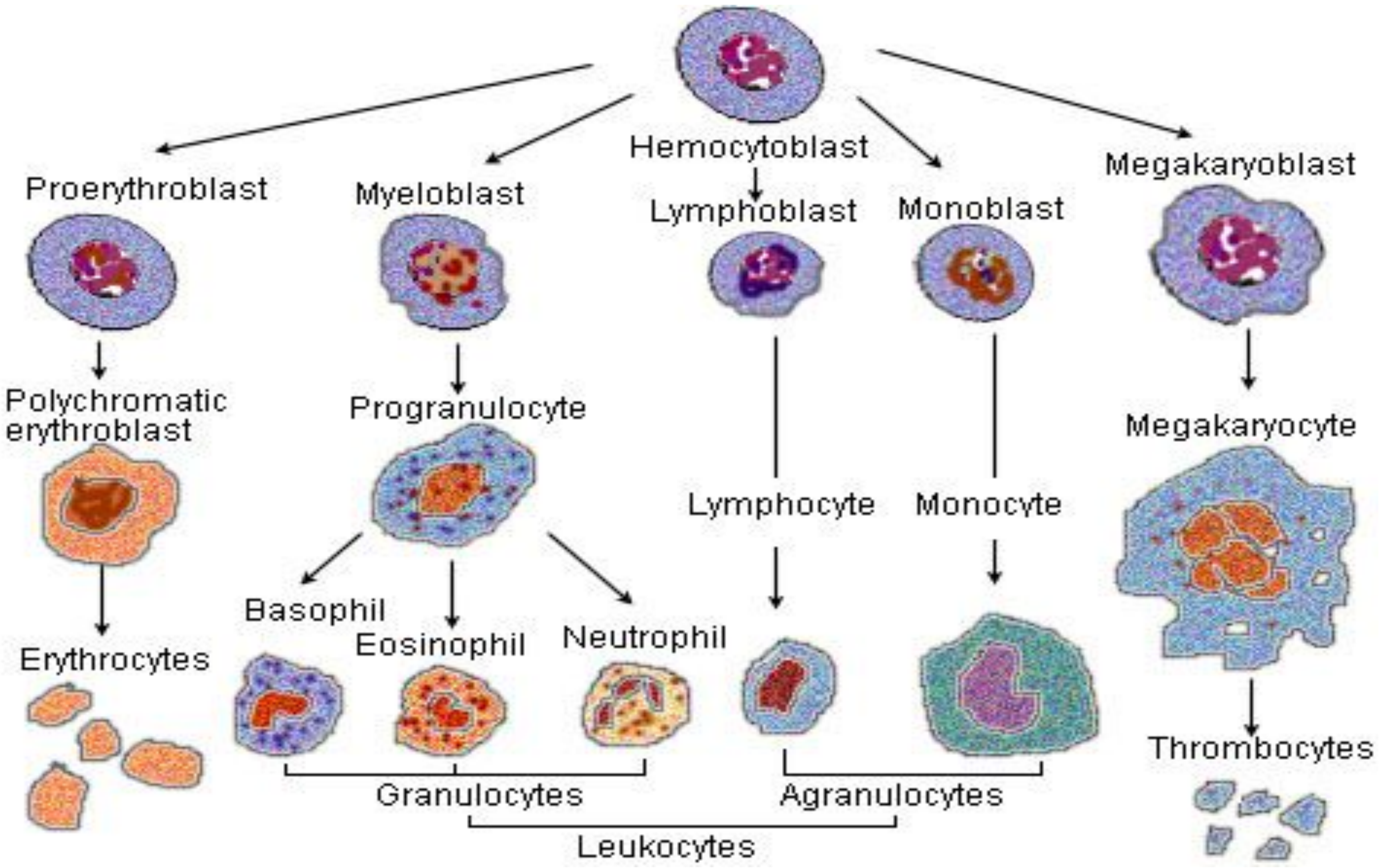
Learning Objectives

- Classify leukocytes into polymorphonuclear granulocytes and the mononuclear agranulocytes.
- Define immunity and describe its types.
- State the defensive function of each type of leukocyte and relate it to the process of inflammation.

Leukocytes White blood cells or WBCs

- Mobile units of body's immune defense system.
- **Immune system**
- Made up of leukocytes, their derivatives, and variety of plasma proteins.
- Recognizes and destroys or neutralizes materials within body that are foreign to "normal self".
- **Functions**
- Defends against invading pathogens.
- Identifies and destroys cancer cells that arise in body.
- Functions as a "cleanup crew" that removes worn-out cells and tissue debris.

Hematopoiesis



WHITE BLOOD CELLS

- **Granulocytes**

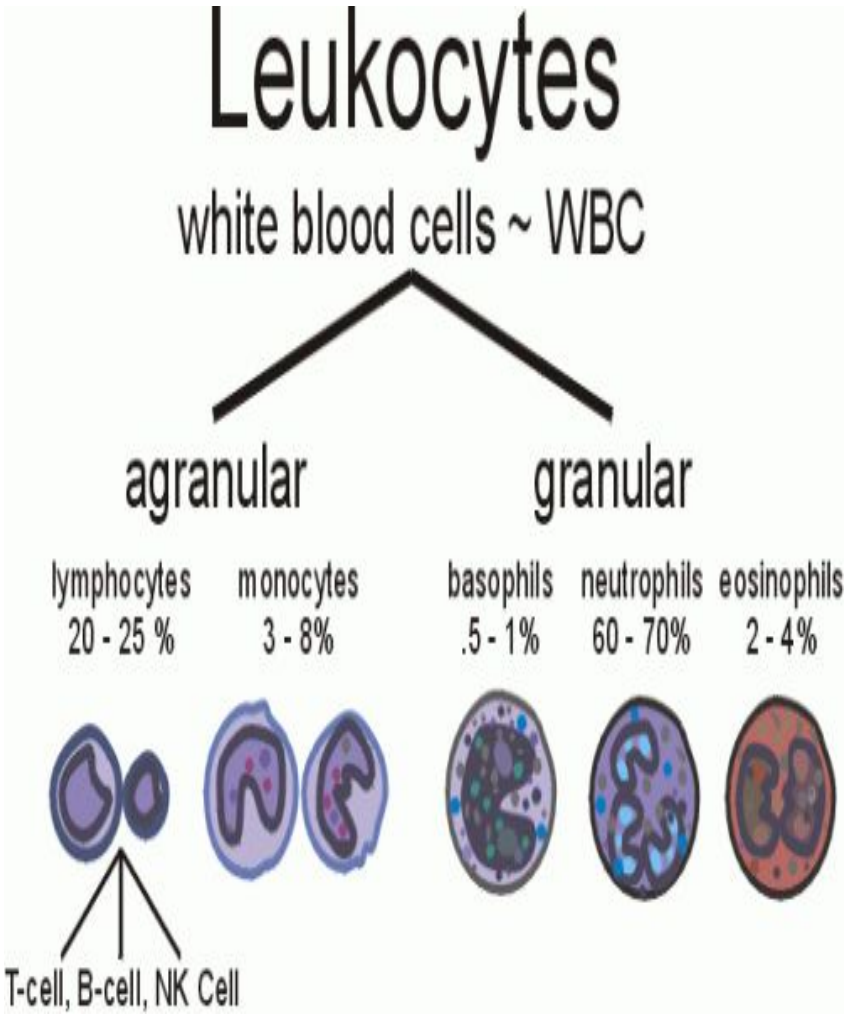
(polymorponuclear leukocytes) include:

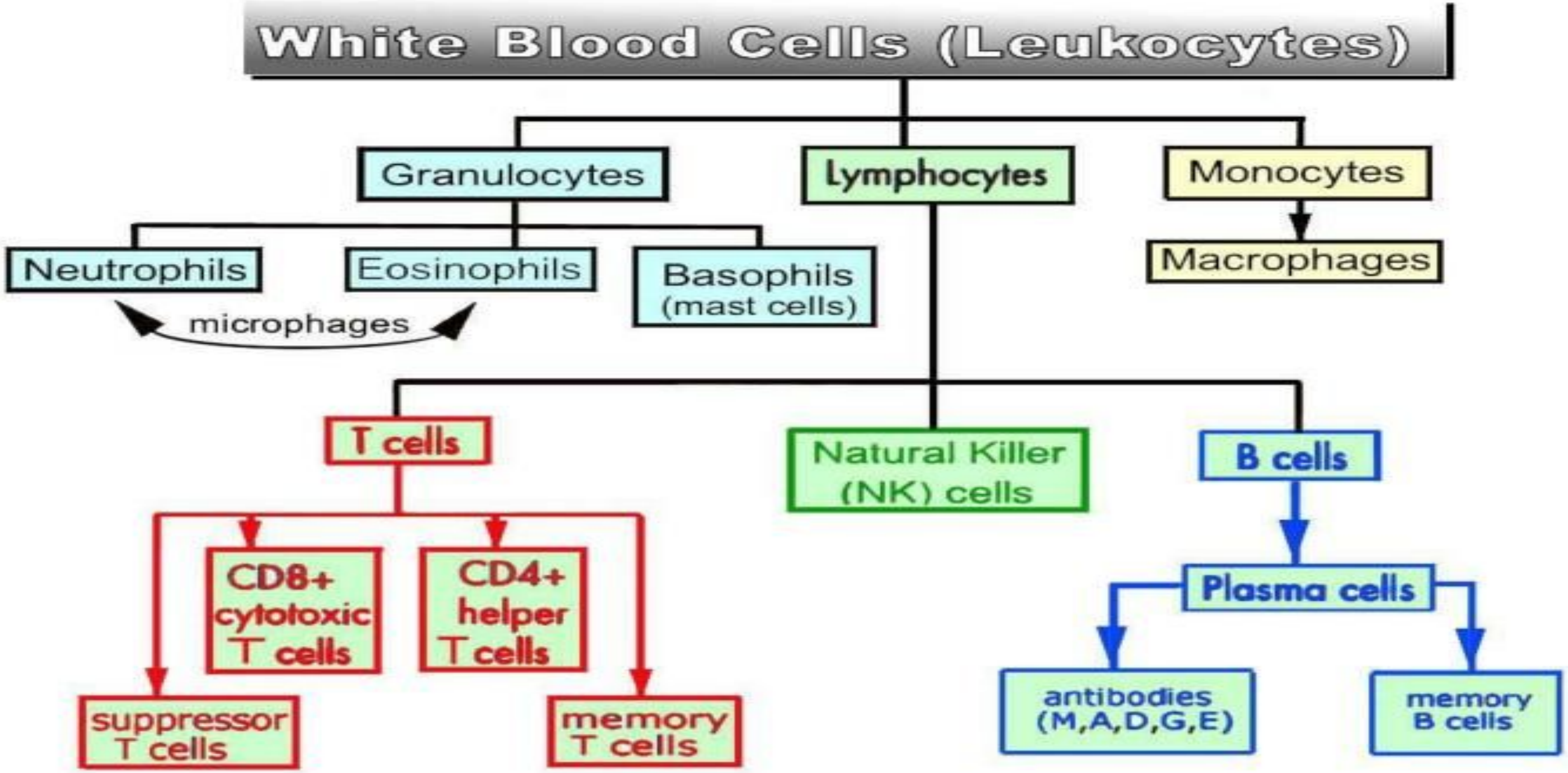
neutrophils, basophils, and eosinophils.

- **Agranulocytes**

(mononuclear leucocytes):

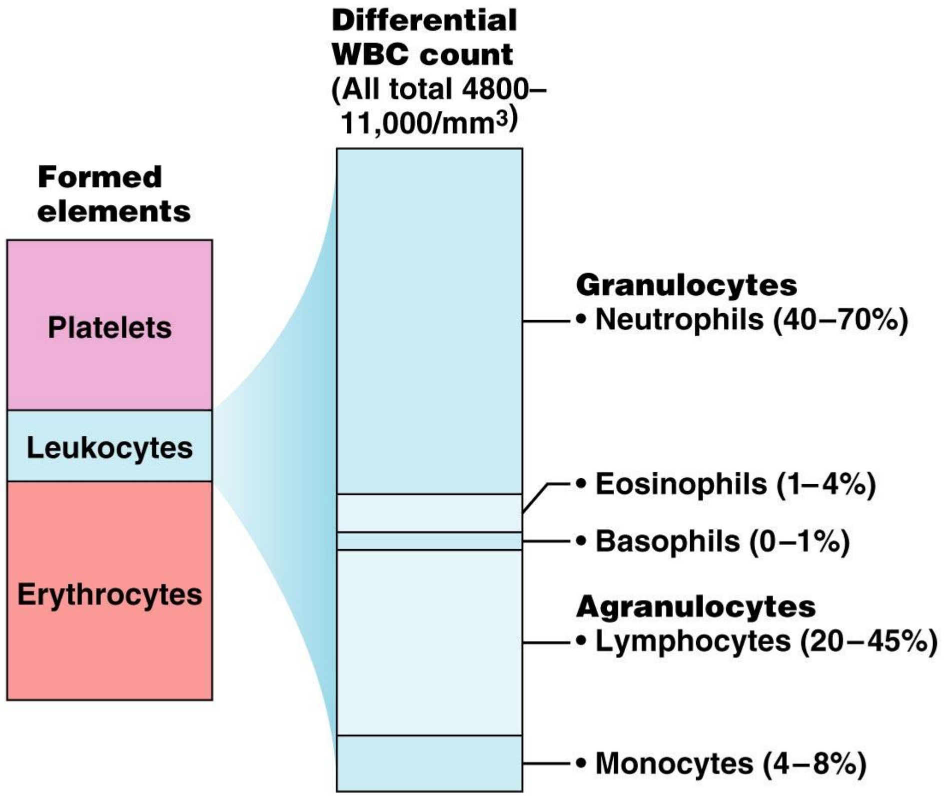
include:
lymphocytes, monocytes, and macrophages.




White Blood Cells (Leukocytes)
Granulocytes
Lymphocytes
Monocytes
Macrophages
Neutrophils
Eosinophils
Basophils
(mast cells)
microphages
T cells
Natural Killer
(NK) cells
B cells
CD8+
cytotoxic
T cells
CD4+
helper
T cells
Plasma cells
suppressor
T cells
memory
T cells
antibodies
(M,A,D,G,E)
memory
B cells

Leukocytes – White Blood Cells (WBCs)

- Two types of leukocytes
  - Granulocytes
  - Agranulocytes
- Differential WBC Count
  - Never
  - Let
  - Monkeys
  - Eat
  - Bananas



Types of white blood cells

TABLE 16.3: Normal values of different WBCs

WBC	Percentage	Absolute value per cu mm
Neutrophils	50 to 70	3,000 to 6,000
Eosinophils	2 to 4	150 to 450
Basophils	0 to 1	0 to 100
Monocytes	2 to 6	200 to 600
Lymphocytes	20 to 30	1,500 to 2,700

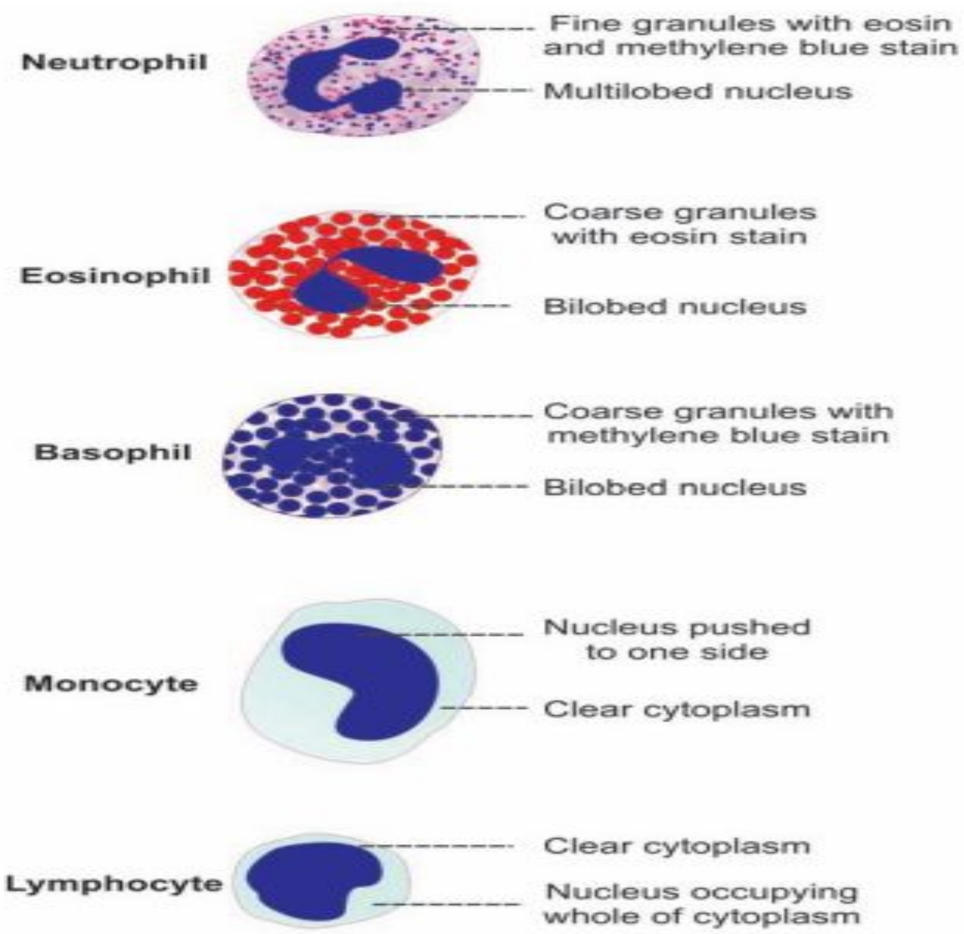


FIGURE 16.1: Different white blood cells

Leucocytes functions

- **Neutrophils** are highly mobile phagocytic specialists that engulf and destroy unwanted materials, defending us from bacteria, fungus etc.
- **Eosinophils** secrete chemicals that destroy parasitic worms.
- **Basophils** release histamine and heparin and also are involved in allergic reactions.
- **Monocytes** are transformed into macrophages, which are large, tissue-bound phagocytic specialists. .
- **Lymphocytes** are of two types: **a**. **B lymphocytes** (B cells) produce antibodies that indirectly lead to the destruction of foreign material (antibody-mediated immunity).**b. T lymphocytes** (T cells) directly destroy virus-invaded cells and mutant cells by releasing chemicals that punch lethal holes in the victim cells (cell-mediated immunity).

Properties of white blood cells

- 1. **Diapedesis** is the process by which the leukocytes squeeze through the narrow blood vessels.
- 2. **Ameboid Movement** Neutrophils, monocytes and lymphocytes show amebic movement, characterized by protrusion of the cytoplasm and change in the shape.
- 3. **Chemotaxis** is the attraction of WBCs towards the injured tissues by the chemical substances released at the site of injury.
- 4**. Phagocytosis** Neutrophils and monocytes engulf the foreign bodies by means of phagocytosis.

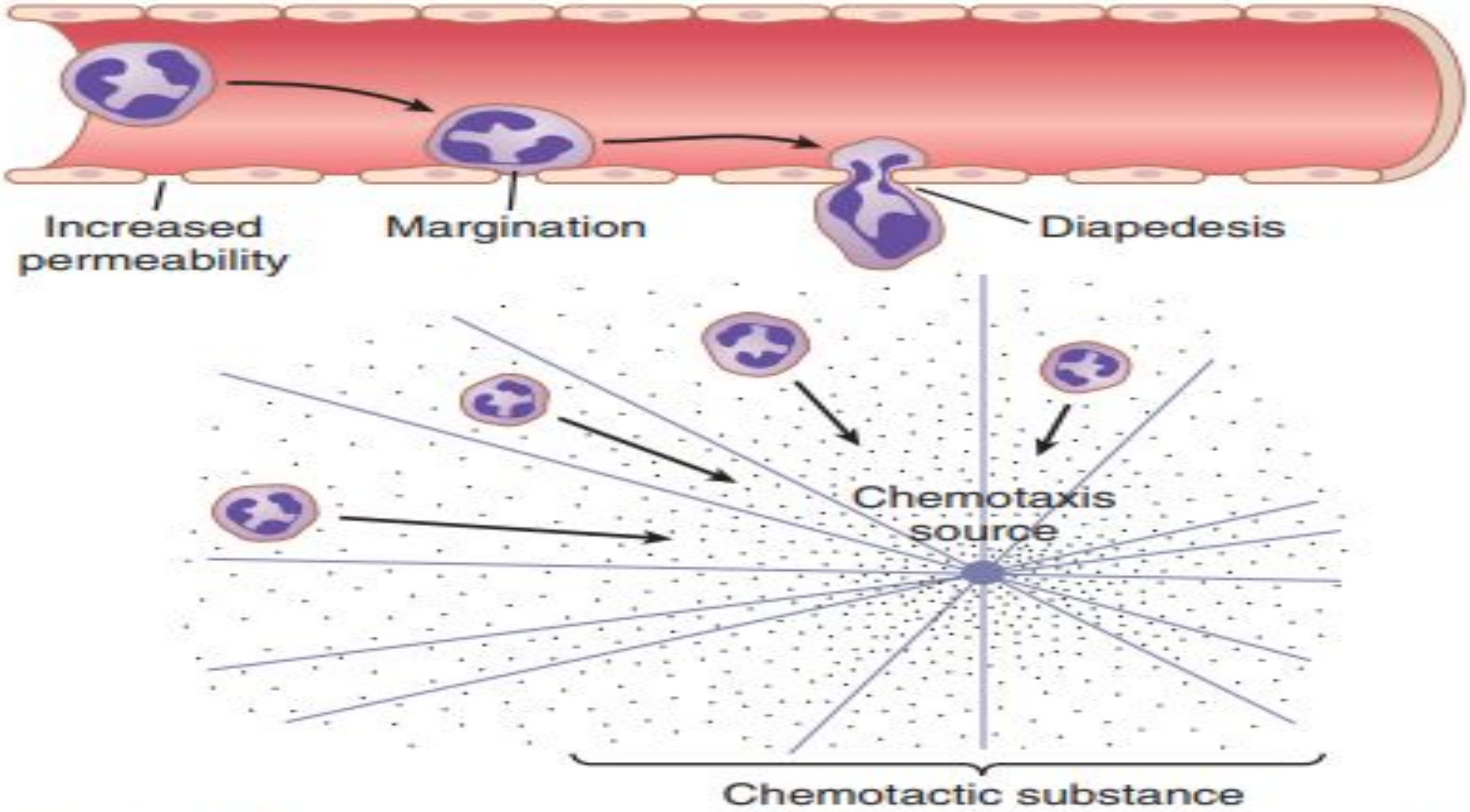
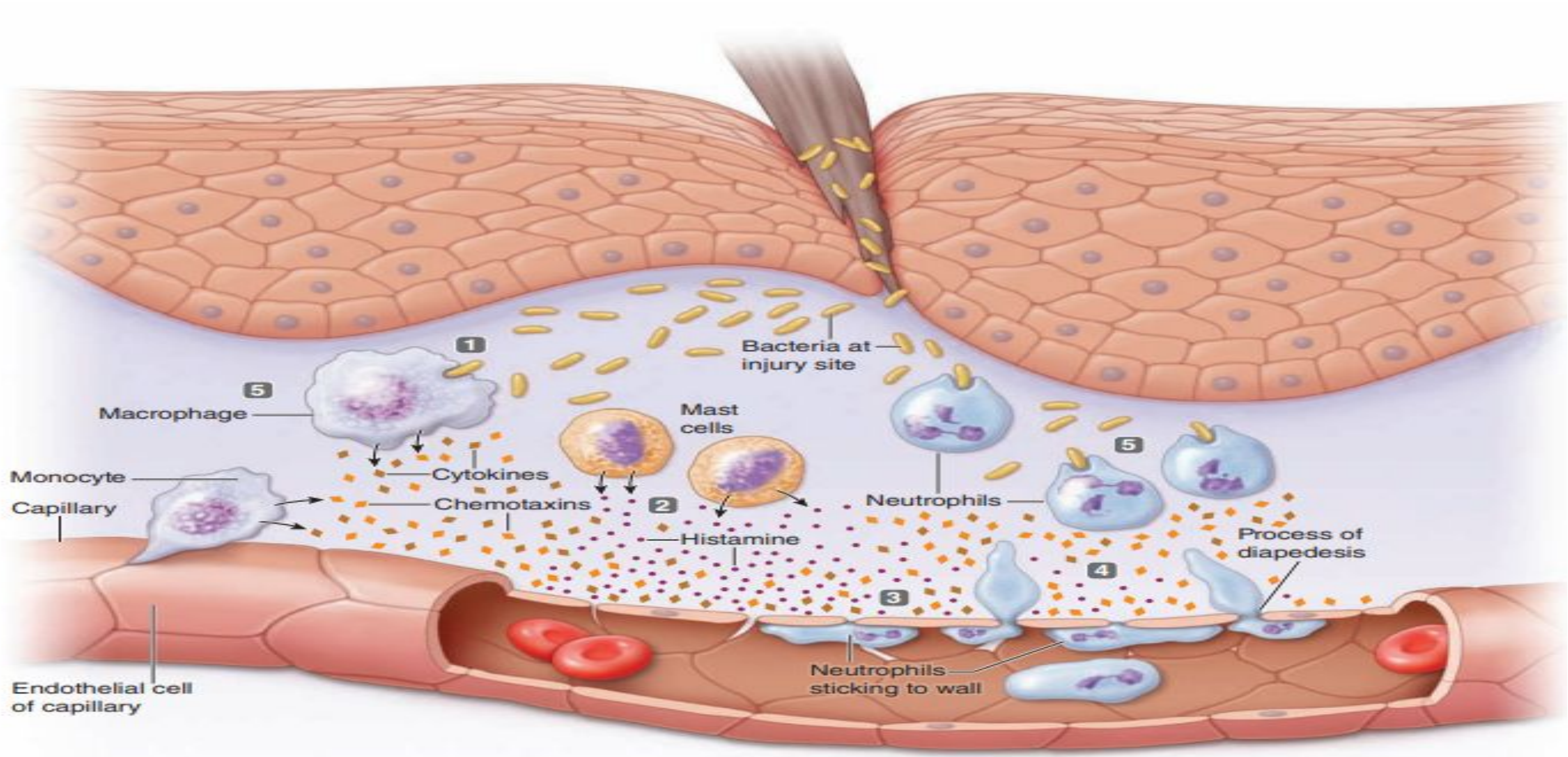


Figure 34-2. Movement of neutrophils by *diapedesis* through capillary pores and by *chemotaxis* toward an area of tissue damage.

Emigration Of Leukocytes

- Within an hour after injury, the area is teeming with leukocytes that have left the vessels. Neutrophils arrive first, followed during the next 8 to 12 hours by the slower-moving monocytes. The latter swell and mature into macrophages during another 8- to 12-hour period. Once neutrophils or monocytes leave the bloodstream, they never recycle back to the blood. Leukocytes can emigrate from the blood into the tissues via the following steps:
- Blood-borne neutrophils and monocytes stick to the inner endothelial lining of capillaries in the affected tissue, a process called **margination** . Selectins, a type of cell adhesion molecule (CAM;) that protrudes from this vessel lining, cause leukocytes flowing by in the blood to slow down and roll along the interior of the vessel, much as the nap of a carpet slows down a child's rolling toy car. This slowing-down allows neutrophils and monocytes enough time to check for local activating factors—"SOS signals", such as cytokines released by resident microphages in nearby infected tissues. When present, these activating factors cause these leukocytes to adhere firmly to the endothelial lining by means of interaction with another type of CAM, the integrins. Soon the adhered leukocytes start leaving by a mechanism known as **diapedesis.** The adhered leucocyte wriggles through the capillary pore & then crawls towards the injured area.

- The phagocytic cells are attracted to chemotaxins, chemical mediators released at the site of damage (taxis means “attraction”). Binding of chemotaxins with protein receptors on the plasma membrane of a phagocytic cell increases Ca entry into the cell. Calcium, in turn, switches on the cellular contractile apparatus that leads to amoeba-like crawling. Because the concentration of chemotaxins progressively increases toward the site of injury, phagocytic cells move unerringly toward this site along a chemotaxin concentration gradient.



1 A break in the skin introduces bacteria, which reproduce at the wound site. Activated resident macrophages engulf the pathogens and secrete cytokines and chemotaxins.

2 Activated mast cells release histamine.

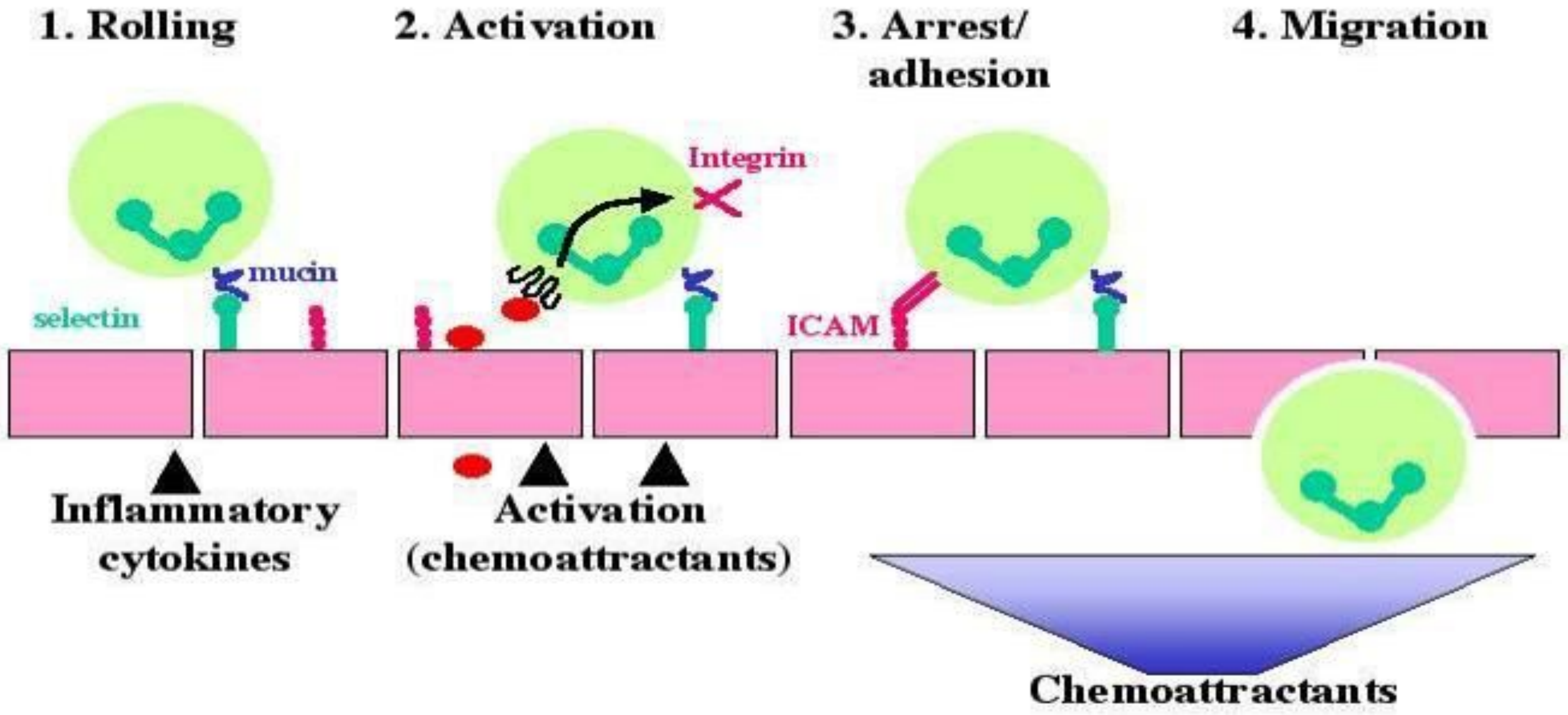
3 Histamine dilates local blood vessels and widens the capillary pores. The cytokines cause neutrophils and monocytes to stick to the blood vessel wall.

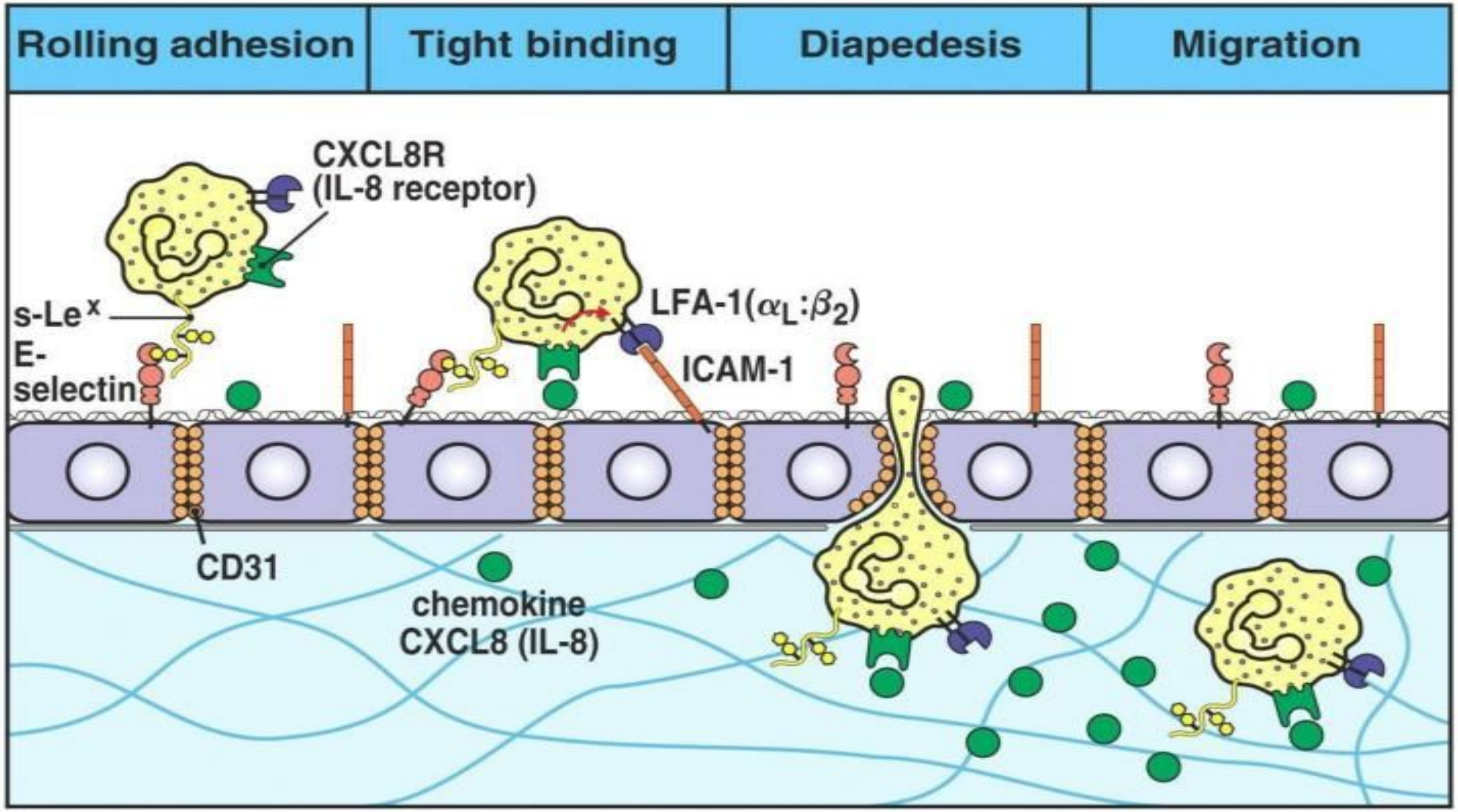
4 Chemotaxins attract neutrophils and monocytes, which squeeze out between cells of the blood vessel wall, a process called diapedesis, and migrate to the infection site.

5 Monocytes enlarge into macrophages. Newly arriving macrophages and neutrophils engulf the pathogens and destroy them.

› **Figure 12-2 Steps producing inflammation.** Chemotaxins released at the site of damage attract phagocytes to the scene. Note the leukocytes emigrating from the blood into the tissues by assuming amoeba-like behavior and squeezing through the capillary pores. Mast cells secrete vessel-dilating, pore-widening histamine. Macrophages secrete cytokines that exert multiple local and systemic effects.

How do Neutrophils get to the site of infection?



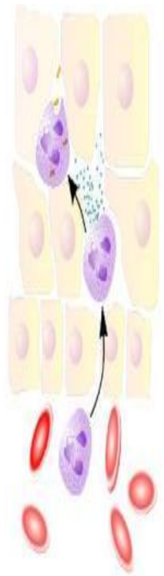

Rolling adhesion
Tight binding
Diapedesis
Migration
CXCL8R
(IL-8 receptor)
s-Le x
E-
selectin
LFA-1(α_L:β_2)
ICAM-1
CD31
chemokine
CXCL8 (IL-8)

Immunity

- **Immunity** is defined as the capacity of the body to resist pathogenic agents. It is the ability of body to resist the entry of different types of foreign bodies like bacteria, virus, toxic substances, etc.
- Immunity is of two types: **I**. Innate immunity. **II.** Acquired immunity
- **Innate or non specific immunity** is the inborn capacity of the body to resist pathogens. By chance, if the organisms enter the body, innate immunity eliminates them before the development of any disease. It is otherwise called the natural or non-specific immunity. This type of immunity represents the first line of defense against any type of pathogens.
- **Acquired or specific immunity** is the resistance developed in the body against any specific foreign body like bacteria, viruses, toxins, vaccines or transplanted tissues. also known as specific immunity.
- Two types of acquired immunity develop in the body: 1. Cellular immunity 2. Humoral immunity. Lymphocytes are responsible for the development of these two types of immunity.

Role of Neutrophils

- Form an essential part of innate immune system.
- recruited to site of injury within minutes following trauma.
- hallmark of acute inflammation.
- Undergo chemotaxis.
- Cell surface receptor allow neutrophils to detect chemical gradients of molecules such as interleukin-8 (IL-8), interferon gamma (IFN-gamma), and C5a, which these cells use to direct the path of their migration.



Neutrophils migrate from blood vessels to the inflamed tissue via chemotaxis, where they remove pathogens through phagocytosis and degranulation

Role of Neutrophils (cont.)

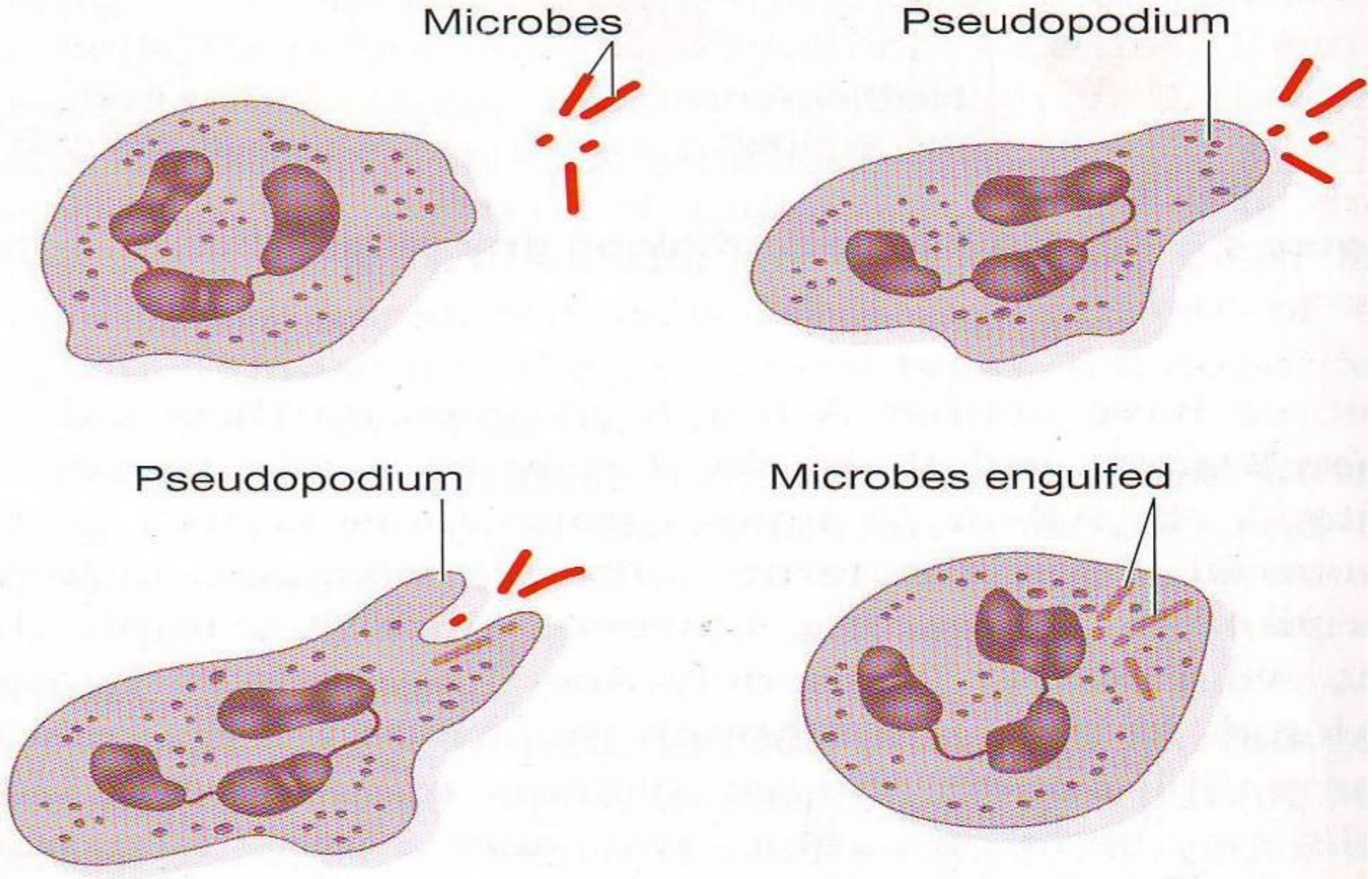
- Being highly motile, neutrophils quickly congregate at a focus of infection, attracted by cytokines expressed by activated endothelium, mast cells, and macrophages.
- Neutrophils express and release cytokines, which in turn amplify inflammatory reactions by several other cell types.
- Have three strategies for directly attacking micro-organisms :

1. Phagocytosis (ingestion).
2. Release of soluble anti-microbials (including granule proteins)
3. Generation of neutrophil extracellular traps (NETs)

Role of Neutrophils (cont.)

PHAGOCYTOSIS:

- Internalize and kill many microbes, each phagocytic event resulting in the formation of a phagosome into which reactive oxygen species and hydrolytic enzymes are secreted.
- The consumption of oxygen during the generation of reactive oxygen species has been termed the "respiratory burst”.
- Respiratory burst is a rapid increase in oxygen consumption during the process of phagocytosis by neutrophils and other phagocytic cells. Nicotinamide adenine dinucleotide phosphate (NADPH) oxidase is responsible for this phenomenon. During respiratory burst, the free radical O_2 – is formed. $2O_2$ – combine with $2H^+$ to form H_2O_2 (hydrogen peroxide). Both O_2 – and H_2O_2 are the oxidants having potent bactericidal action.



Phagocytic action of neutrophils.

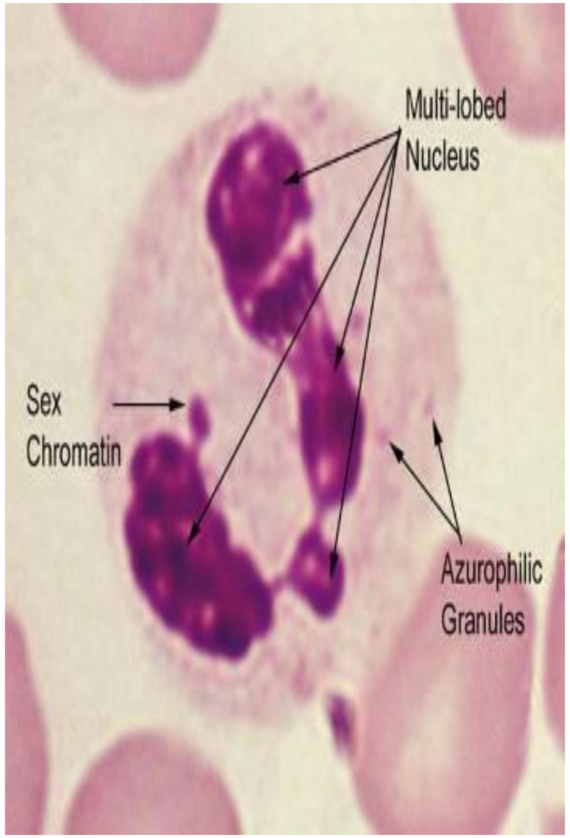
Role of Neutrophils (cont.)

Degranulation:

- Neutrophils also release an assortment of proteins in three types of granules by a process called degranulation.

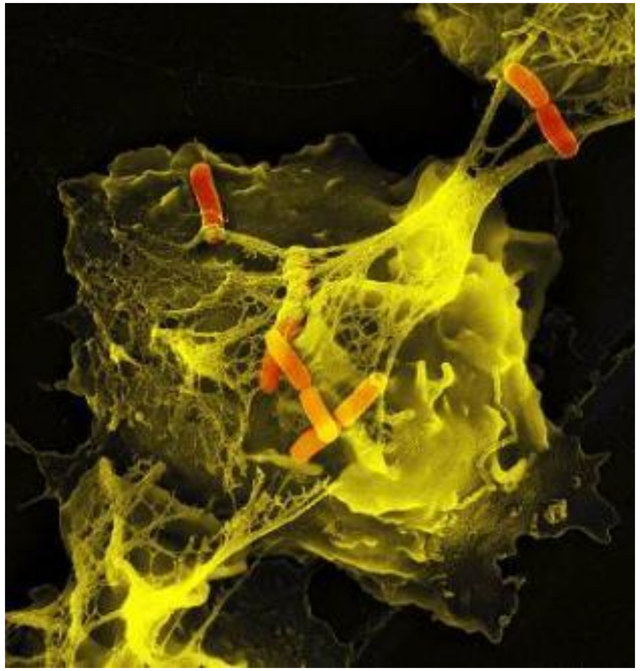
Have three types of granules:

1. Specific granules (or "secondary granules").
2. Azurophilic granules (or "primary granules").
3. Tertiary granules.



Neutrophil Extracellular Traps Nets

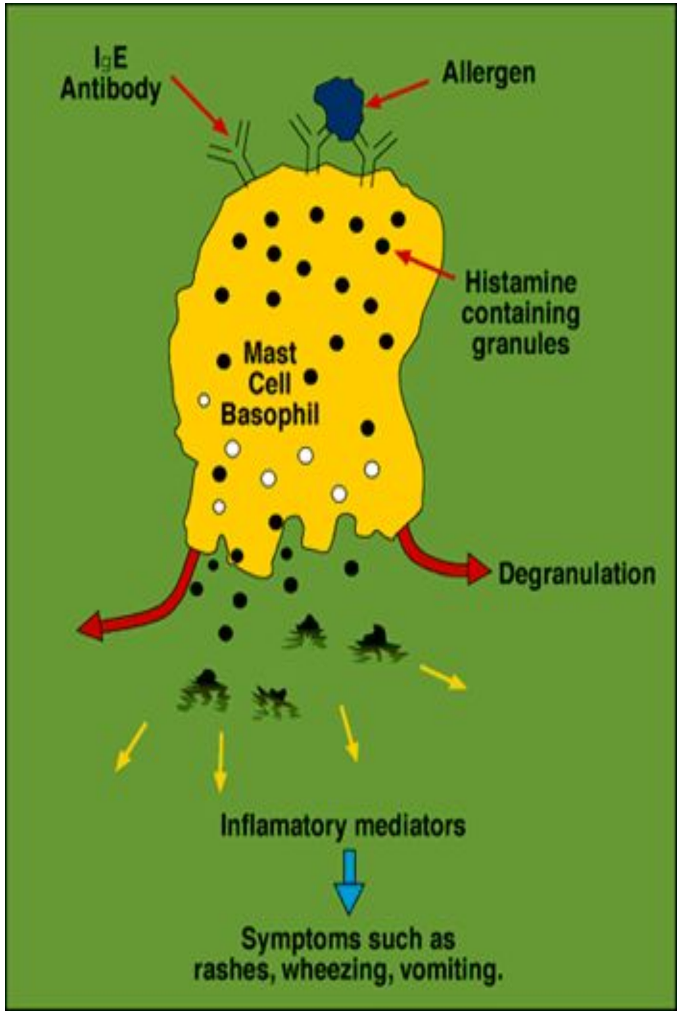
- Activation of neutrophils causes the release of web-like structures of DNA !
- These neutrophil extracellular traps (NETs) comprise a web of fibers composed of chromatin and serine proteases that trap and kill microbes extracellularly .
- Recently discovered mechanism of neutrophil killing.



Stimulated neutrophil with NETs and some trapped Shigella (orange)

Role Of Basophils

- Basophils appear in specific kinds of inflammatory reactions, particularly those that cause allergic symptoms
- Basophils contain anticoagulant heparin.
- contain vasodilator histamine.
- They can be found in unusually high numbers at sites of ectoparasite infection, e.g., ticks.
- Like eosinophils, basophils play a role in both parasitic infections and allergies
- Basophils have protein receptors on their cell surface that bind IgE, an immunoglobulin involved in macroparasite defense and allergy.



Role Of Eosinophils

- Eosinophils are weak phagocytes, and they exhibit chemotaxis.
- Eosinophils, are often produced in large numbers in people with parasitic infections, and they migrate into tissues diseased by parasites.
- Eosinophils attach themselves to the juvenile forms of the parasite and kill many of them. by: (1) by releasing hydrolytic enzymes from their granules, which are modified lysosomes; (2) probably by also releasing highly reactive forms of oxygen that are especially lethal to parasites; and (3) by releasing from the granules a highly larvacidal polypeptide called major basic protein.
- The mast cells and basophils release an eosinophil chemotactic factor that causes eosinophils to migrate toward the inflamed allergic tissue.

Role Of Monocytes

- Monocytes are the largest cells among the leukocytes. Monocytes have two main functions in the immune system:

(1) Replenish resident macrophages and dendritic cells under normal states

(2) In response to inflammation signals, monocytes can move quickly (approx. 8-12 hours) to sites of infection in the tissues and divide/differentiate into macrophages and dendritic cells to elicit an immune response.

Monocytes play an important role in defense of the body. Along with neutrophils, these leukocytes provide the first line of defense. Monocytes secrete: 1. Interleukin-1 (IL-1). 2. Colony stimulating factor (M-CSF). 3. Platelet-activating factor (PAF). Monocytes are the precursors of the tissue macrophages. Matured monocytes stay in the blood only for few hours. Afterwards, these cells enter the tissues from the blood and become tissue macrophages. Examples of tissue macrophages are Kupffer cells in liver, alveolar macrophages in lungs and macrophages in spleen.

Role Of Macrophages

Three important roles :

1.Phagocytosis.

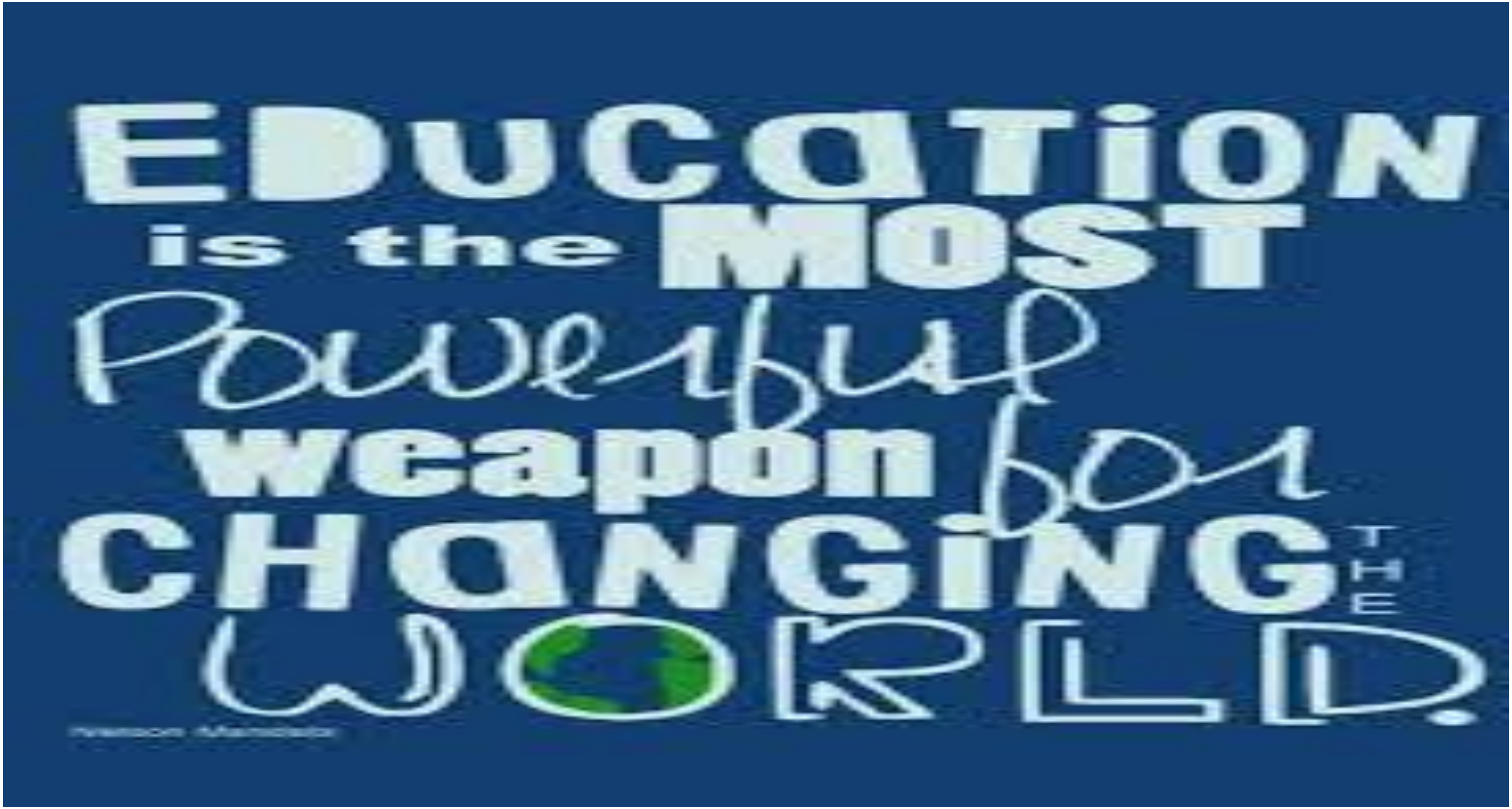
2.Antigen presentation.

3.Muscle regeneration.

Macrophage destroying bacterial cells

Functions of tissue macrophages:

- **1.** Phagocytic Function
- **2**. Secretion of Bactericidal Agents
- **3**. Secretion of Interleukins Tissue macrophages secrete the following interleukins, which help in immunity: i. Interleukin-1 (IL-1): Accelerates the maturation and proliferation of specific B lymphocytes and T lymphocytes. ii. Interleukin-6 (IL-6): Causes the growth of B lymphocytes and production of antibodies. iii. Interleukin-12 (IL-12): Influences the T helper cells
- **4.** Secretion of Tumor Necrosis Factors Two types of tumor necrosis factors (TNF) are secreted by tissue macrophages**: i.** TNF-α: Causes necrosis of tumor and activates the immune responses in the body **ii.** TNF-β: Stimulates immune system and vascular response, in addition to causing necrosis of tumor.
- **5.**Secretion of Transforming Growth Factor
- **6.**Secretion of Colony-stimulation Factor Colony-stimulation factor (CSF) secreted by macrophages is M-CSF. It accelerates the growth of granulocytes, monocytes and macrophages.
- **7**. Secretion of Platelet-derived Growth Factor Tissue macrophages secrete the platelet-derived growth factor (PDGF), which accelerates repair of damaged blood vessel and wound healing.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